

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORKING SCANNING PHASES

WEEK 2-PM-FINAL REPORT | CYBERSECURITY | NETWORKWALKS

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Batch	B083
Date	18 August 2026
Modules completed	W2-PM1 (KALI LINUX TOOLS) W2-PM4 (THE HARVESTER) W2-PM5 (ZENMAP SCANNING)
Client/Target	1. Networkwalks(secured written permission already) 2. My own LAN Network
Permission secured from client?	yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. INTRODUCTION

This Week 2 report documents my hands-on cybersecurity activities as part of my ongoing internship program at NetworkWalks. The work focuses on two important stages of network security assessment: footprinting and network scanning. In the first module, I used various Kali Linux tools to perform footprinting of the "networkwalks.com" domain and gather publicly available information about the target. In the second module, I used Zenmap on my Windows PC to scan my own local network and identify active hosts and network details.

Together, these activities helped me understand how information gathering and network scanning are used during the initial stages of a security assessment. I also documented the commands used, observed results, screenshots, and the security relevance of each activity.

All activities were performed in authorized internship/lab environment for learning and cybersecurity practice.

3. Tools Used

The table below list each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for reconnaissance activities
Whois	Find domain registration details (owner, dates, name servers).
Whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
Wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV)
The Harvester	OSINT tool for gathering emails, subdomains and hosts related to a domain from public sources
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses

4. Activities performed

4.1 Footprinting & Reconnaissance with Kali Tools (W2-PM1)

I performed authorized reconnaissance of the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon. Each tool was used to collect different types of publicly available information about the domain.

WHOIS provided domain registration and name server details. WhatWeb identified technologies used by the website. The results identified WordPress 7.1 and WP Download Manager 3.3.58 along with the other information exposed by the website.

Nslookup helped resolve the domain to its IP address (192.232.216.135). Curl was used to inspect HTTP response headers. Wafw00f detected the presence of ModSecurity (SpiderLabs), and DNSRecon helped enumerate DNS records such as name servers and mail-related records.

This activity helped me understand the footprinting and reconnaissance phase of cybersecurity and how different Kali Linux tools can be combined to gather information about a target.

4.2 TheHarvester based Footprinting Attacks (W2-PM4)

I reviewed theHarvester (v4.10.1) and its available options for gathering open-source intelligence on a company or domain. The tool supports specifying a target domain (-d), limiting results (-l), using proxies (-p) or Shodan (-s), taking screenshots of resolved domains (--screenshot), specifying a DNS server (-e), checking for subdomain takeover (-t), and resolving discovered hosts to IP addresses (-r), among other options. This confirmed the breadth of OSINT sources (search engines, DNS, Shodan) that can be combined to build a picture of an organisation's external footprint before any active engagement begins.

4.3 Network Scanning with Zenmap (W2-PM5)

For this activity, I used Zenmap to perform network discovery on my local network. I first identified my local IP address and subnet using the Windows ipconfig command. The identified network was 10.175.188.0/24.

I then entered the subnet into Zenmap and performed a Ping Scan (nmap -sn) to identify active devices on the network. The scan detected four live hosts: 192.168.1.2, 192.168.1.3, 192.168.1.5, 192.168.1.7

After completing the scan, I checked the discovered host information and opened the Topology tab in Zenmap. The topology view displayed the detected hosts and their connections with my local system. I also enabled the topology legend and saved the network topology as a PDF as required for the practical task.

This activity helped me understand how Zenmap can be used for host discovery, network mapping, and basic network visualization.

5. Risk analysis/Impact

#	Risk /Finding	Evidence/ Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use hexposed technology/version information to identify software requiring further security review	● Medium

2	Server IP address identifiable	Nslookup resolved the domain to (192.232.216.135)	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	OSINT tooling can enumerate further organisational footprint	The Harvester supports domain, DNS-resolve, take-over-check and Shodan-based enumeration	Indicates the breadth of public information an attacker could gather with minimal effort	● Medium
7	Multiple live hosts visible on local network	Zenmap identified four live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these three modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Monitor publicly available technology details

Organizations should regularly check what information about their websites, CMS platforms, plugins and technologies is visible to the public.

2. Keep web technologies updated

CMS platforms, plugins and other software should be updated regularly to reduce security risks and address known vulnerabilities.

3. Check HTTP security headers

HTTP response headers should be reviewed to identify unnecessary technical details and improve the security configuration.

4. Regularly review DNS information

DNS records should be checked periodically to make sure only necessary services and information are exposed publicly.

5. Configure and monitor the WAF properly

The Web Application Firewall (ModSecurity) should remain enabled and properly configured to help block common web-based attacks.

6. Conduct regular network discovery

Organizations should periodically scan their authorized internal networks to identify active devices and understand the network environment.

7. Verify unknown or unexpected devices

Any unfamiliar device found during network scanning should be investigated and confirmed as an authorized system.

8. Keep network documentation updated

Network topology, IP addresses and device details should be properly documented and updated whenever changes are made.

9. Perform security testing only with authorization

Reconnaissance, scanning and other security testing activities should only be performed on systems and networks for which proper permission has been obtained.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the theHarvester activity, I reviewed how a dedicated OSINT tool combines multiple public sources (search engines, DNS, Shodan) into one workflow.

Across these three modules, the exercises showed me that information gathering is a foundational part of cybersecurity, and that a significant amount can be learned

Finally, I learned that reconnaissance and scanning must always be performed within an authorised scope. These activities were completed as part of the assigned educational cybersecurity lab.


```

$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp/clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of Whois database: 2026-09-16T11:31:24Z <<<

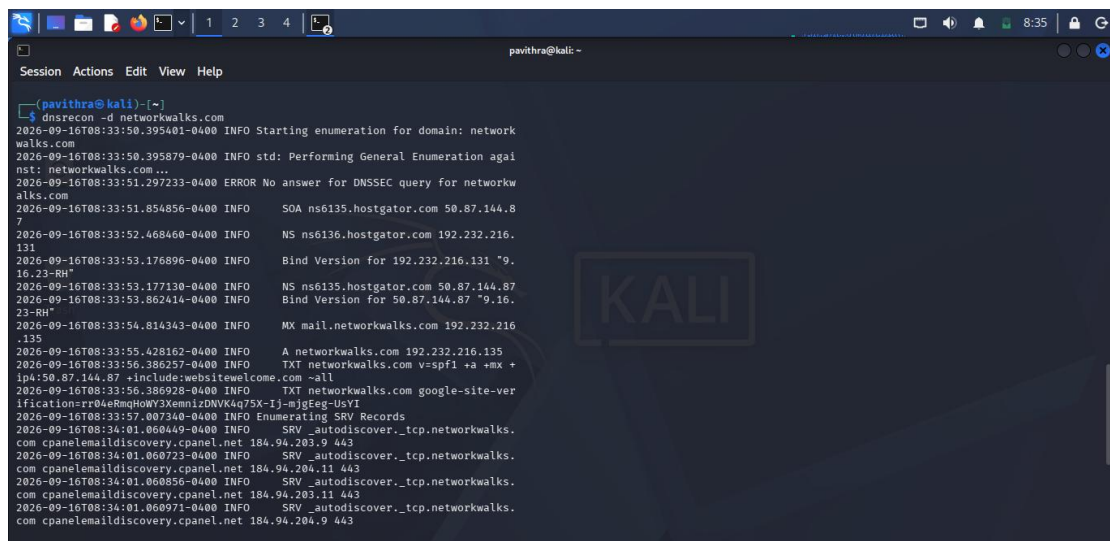
For more information on Whois status codes, please visit https://icann.org/epp

NOTICES: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and

```

6



```

/home/pavithra

Home
File System
Trash

pavithra@kali: ~
Session Actions Edit View Help
(pavithra@kali)-[~]
└─$ curl -I https://networkwalks.com
curl: (6) Could not resolve host: networkwalks.com

(pavithra@kali)-[~]
└─$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?; rel="https://api.w.org/">, <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/?; rel=shortlink
set-cookie: __wpdm_client=32b7116dee75e531eb10c3c191484998; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Wed, 16 Sep 2026 12:07:29 GMT
server: Apache

(pavithra@kali)-[~]
└─$
```

```

Text Editor
Simple Text Editor Help

└─$ theHarvester -d microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] Emails found: 4
,contact@microsoft.com
support@microsoft.com
info@microsoft.com
viva-noreply@microsoft.com

[*] No people found.

[*] Hosts found: 1
officecdn.microsoft.com
```

```

/home/pavithra

pavithra@kali: ~
Session Actions Edit View Help
(pavithra@kali)-[~]
└─$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
```

